THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105 Environmental Engineering

May 14, 2014 TIME: 2:30 – 5:30 pm

I. Answer ALL questions. Section A carries 60 marks, and Section B carries 40 marks.

II. Atomic weights: H 1; C 12; F 19; O 16; Na 23; Mg 24.3; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (20 multiple choice questions with 3 marks each)

1. Which of the following contaminants in wastewater is not easily removed by the combination of primary clarification and secondary biological treatment? (a) BOD, (b) suspended solids, (c) biodegradable organic matter, (d) phosphorous, or (e) particles with diameter > 10 μm.
2. A factory has a machine that operates at 80 dB. The factory plans to purchase two additional machines that will operate at 76 dB and 90 dB, respectively. What will be the combined noise level from the three machines? (a) 76 dB, (b) 80 dB, (c) 82 dB, (d) 85 dB, (e) 90 dB, (f) 170 dB, or (g) 246 dB?
3. Which of the following process is not suitable for hardness removal? (a) Nanofiltration, (b) lime-soda method, (c) sedimentation, or (d) ion exchange.
4. Which of the following chemicals is the weakest disinfectant? (a) Chlorine, (b) ozone, (c) chlorine dioxide, or (d) chloramine.
5. You are asked to provide a rough estimation of the density of air at sea level. Which is the closest value? (a) 1.2 g/m^3, (b) 12 g/m^3, (c) 120 g/m^3, (d) 1200 g/m^3, or (e) 12000 g/m^3.
6. Which of the following compounds does not cause greenhouse effect? (a) Water vapor, (b) carbon dioxide, (c) methane, (d) halocarbons, or (e) nitrogen?
7. Which of the following air pollutants is not included in determining the Air Quality Health Index (AQHI) in Hong Kong? (a) CO, (b) ground-level ozone, (c) NO_2, (d) SO_2, and (e) particulates.
8. Which of the following air pollutants is not treated by a three-way catalytic converter? (a) NO, (b) CO, (c) SO_2, or (d) hydrocarbons.
9. The ambient air temperature at the ground level is 26 °C. As the altitude increases, the air temperature first decreases at 15 °C/km for the first 400 m (i.e., temperature at 400 m above the ground level is 20 °C) before it increases at 2.5 °C/km. The air is dry so that the dry adiabatic lapse rate is applicable. What is the mixing depth? (a) 200 m, (b) 300 m, (c) 400 m, (d) 480 m, (e) 560 m, or (f) 640 m.
10. Which of the following phenomena is not directly related to increased CO_2 emission? (a) Ocean acidification, (b) acid rain, (c) global warming, (d) raising sea level, or (f) accelerated melting of glaciers.
11. A student puts some limestone ($CaCO_3(s)$) to a beaker of water. A small portion of the added limestone is dissolved, and the rest remains as solid. What will happen to the solution if carbon dioxide (CO_2) is bubbled into the beaker? (a) pH will decrease and calcium concentration will decrease, (b) pH will decrease and calcium concentration will increase, (c) pH will increase and calcium concentration will decrease, or (d) pH will increase and calcium concentration will increase.
12. What is the theoretical total oxygen demand of a wastewater sample containing 100 mg/L $C_4H_9NO_2$? (a) 50 mg/L, (b) 60 mg/L, (c) 101 mg/L, (d) 140 mg/L, or (e) 202 mg/L.
13. The half life of radon (Rn-222) is 3.8 days. If the initial mass of Rn-222 is 1 g, how much will be remaining after 1 week? (a) 0.14 g, (b) 0.16 g, (c) 0.28 g, (d) 0.46 g, or (e) 0.54 g.
14. An ultrafiltration membrane is able to remove 99.9999% of microorganisms from a feed water. What is the log-removal of the membrane? (a) 0.999999, (b) 2, (c) 4, (d) 6, or (e) 999999.
15. In an anaerobic digester, organic matter in the sludge would normally go through the transformation of: (a) complex organic matter → hydrolyzed small organic molecules → volatile organic acids → methane; (b) complex organic matter → hydrolyzed small organic molecules → methane → volatile organic acids; (c) complex organic matter → volatile organic acids → hydrolyzed small organic molecules → methane; (d) complex

organic matter → volatile organic acids → methane → hydrolyzed small organic molecules; or (e) cannot tell due to insufficient information.

16. A small town plans to treat a river water for its drinking water supply. Which of the following treatment sequence is most appropriate? (a) disinfection → sedimentation → flocculation → coagulation → filtration, (b) sedimentation → flocculation → coagulation → filtration → disinfection, (c) sedimentation → coagulation → flocculation → filtration → disinfection, (d) sedimentation → flocculation → coagulation → disinfection → filtration, or (e) sedimentation → coagulation → flocculation → disinfection → filtration.
17. During hot summer months, water in the upper layer of a lake is warmer than that in the bottom layer. This leads to a density difference between the layers, which inhibits the vertical mixing of the lake. The above phenomenon is known as (a) temperature inversion, (b) density inversion, (c) thermal stratification, or (d) nitrification.
18. California's air quality standard for CO is 20 ppm (1-hour average). According to the ideal gas law, one mole of ideal gas occupies 24.5 L at 25 °C and 1 atm. Express the concentration level of CO in mg/m^3: (a) 11 mg/m^3, (b) 16 mg/m^3, (c) 23 mg/m^3, (d) 28 mg/m^3, or (e) 35 mg/m^3.
19. Which of the following biological treatment belongs to the attached growth treatment? (a) activated sludge, (b) aerated lagoon, (c) membrane bioreactor, (d) rotating biological contactor, or (e) oxidation pond.
20. Which process is most suitable for removing SO_2 from power plant flue gas? (a) electrostatic precipitator, (b) cyclone, (c) baghouse, (d) selective catalytic reduction, or (e) none of the above.

Section B (Two questions with 20 marks each)

21. A city has a rectangular shape with a width of 10 km and a length of 15 km. There are 500,000 cars in the city. On average, each car has a daily driving distance of 40 km, and it emits 0.2 g/km of NO_x.

(a) Estimate the NO_x emission rate per unit area (in unit of $g/s.m^2$).

(3 marks)

(b) The NO_x concentration in the city can be estimated using a box model. What other important information is required for the model? Derive an expression for the steady-state NO_x concentration using the mass balance principle. Is this concentration representative of the peak road side NO_x concentration?

(11 marks)

(c) Briefly explain the phenomenon of photochemical smog. How is it related to NO_x?

(6 marks)

22. (a) BOD and COD are important indicators of the organic strength of a wastewater. Briefly discuss the difference of the two parameters. What is the significance of BOD/COD ratio for wastewater treatment?

(7 marks)

(b) In a 5-day BOD test, a 5 ml wastewater sample is added to a 300 ml BOD bottle. The initial DO is 9 mg/L and the final DO is 4 mg/L. The rate constant is 0.2 day^{-1}. What is the ultimate BOD?

(4 marks)

(c) The Streeter-Phelps oxygen sag curve is widely used to model the effect of wastewater discharge on the dissolved oxygen concentration in a river. Sketch an oxygen sag curve and briefly explain the major mechanisms involved. How does increasing temperature affect the oxygen sag curve?

(9 Marks)

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